

Digital Habits Survey Analysis

An end-to-end analytics pipeline — from Google Forms survey collection through Python cleaning, MySQL/SQL analysis, to an interactive 2-page Looker Studio dashboard exploring digital usage, screen time, and productivity.

ROLE	TYPE	TOOLS	DATASET
Data Analyst / Project Owner	Self-Initiated, End-to-End	Google Forms · Excel · Python · MySQL · SQL · Sheets · Looker Studio	68 self-collected respondents (convenience sample)

68 RESPONDENTS	3.32 AVG. PRODUCTIVITY	45.59% EXCESSIVE SCREEN TIME	~45 SQL QUERIES	7 TOOLS END-TO-END
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Pipeline: Google Forms → Raw Export → Excel/CSV → Python Cleaning → MySQL → SQL Analysis (CTEs + Window Fns) → Google Sheets → Looker Studio Dashboard

DATA MODEL

KEY FINDINGS

SKILLS DEMONSTRATED

Column	Description
age_group, gender, occupation	Respondent demographics
screen_time	Daily screen-time bracket
most_used_platform	Primary digital platform
online_learning	Online learning use (Y/N)
screen_time_excessive	Self-view on excessive use
productive	Productivity score (INT)

- **Instagram dominates** — 39 of 68 respondents (57%), far ahead of any other platform.
- **Working professionals** make up 52 of 68 respondents, skewing occupation comparisons.
- Sample skews heavily to **ages 21-30** (60 of 68).
- Screen time is fairly spread: 2-4 hrs (18) and >8 hrs (16) are the largest brackets.
- Freelancers show the **highest average productivity** (3.5) — but n=2, so unstable.
- Respondents reporting **>8 hrs** screen time show highest observed productivity (3.81).
- Self-identified **excessive screen time** correlates with higher productivity (3.48 vs. 3.05) — association, not causation.

SQLCTEsWindow Fns

RANK()MySQLPython

Data Cleaning

Data ValidationExcel

Google SheetsLooker Studio

Dashboard Design

KPI DesignSurvey Design

Interactive Filtering

Insight Generation

Troubleshooting

SAMPLE SQL — CTE + WINDOW FN

```
WITH occupation_productivity AS (  
  SELECT occupation, AVG(productive) AS avg_p  
  FROM digital_habits_survey  
  GROUP BY occupation  
)  
SELECT occupation, avg_p,  
  RANK() OVER (ORDER BY avg_p DESC) AS rnk  
FROM occupation_productivity;
```

CHALLENGES & FIXES

- **Encoding bug:** UTF-8/Latin-1 mis-encoding corrupted dash characters ("21-30" → "21â€"30") on import; identified in MySQL and corrected via string replacement.
- **MySQL Error 1175:** Safe update mode blocked an UPDATE lacking a key-based WHERE clause — resolved by scoping the update properly.
- **Looker refresh quirks:** Data preview issues when adding calculated fields — resolved through standard source/report iteration.

DASHBOARD — 2 PAGES

- **Page 1 (Overview):** 4 KPI cards + charts for age group, occupation, platform usage, and screen-time distribution.
- **Page 2 (Productivity):** productivity vs. occupation, screen time, online learning & excessive-screen-time perception, plus 3 interactive filters (Gender, Age Group, Occupation).

VERIFICATION FLAGS

- Online Learning KPI (64.71%) vs. donut chart (51.3% Yes) reflect two different calculations — flagged, not yet reconciled.
- Gender distribution exists only as a Page 2 filter, not a chart — unverified breakdown.
- Full Python transformation logic and complete Q1-Q45 SQL list to be confirmed against source files before publishing.

WHY THIS PROJECT

Mirrors a real analyst workflow: ambiguous raw data → cleaning → structured querying → visual storytelling → defensible, well-scoped insights — including documenting unresolved discrepancies rather than papering over them.

Observational, self-collected survey — findings describe patterns within this sample, not causal or generalizable claims. Small sub-samples (Freelancer n=2, Other n=4) are flagged accordingly.